

Report 12 ML Pipeline Fishing Suitability Model

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Report 12 — ML Pipeline: Fishing Suitability Model & Data Science Architecture

Project: JalJeev — Agentic Marine Intelligence System
Phase: Phase 0-1
Focus: XGBoost suitability model, feature engineering, training pipeline, the PFZ label dependency

1. Objective

Build the machine learning pipeline that will power the Fishing Agent — predicting fishing zone suitability from oceanographic features. This report documents the model architecture, feature engineering pipeline, and the critical data dependency that blocks deployment.

2. Model Specification

2.1 Task Definition

Attribute	Value
Task	Binary classification (suitable / not suitable for fishing)
Model	XGBoost (gradient boosted trees)
Alternative	LightGBM (for ensemble/A/B comparison)
Input	Tabular features derived from ocean/weather/geo state
Output	Probability of fishing suitability (0.0 - 1.0)
Training labels	INCOIS PFZ advisory archive (2003-2025)

2.2 Why XGBoost (Not a Neural Network)

- **Tabular data** — XGBoost consistently outperforms deep learning on structured tabular features
 - **Interpretability** — feature importance is directly available, essential for explainability
 - **Training speed** — trains in minutes, not hours; iterates fast during development
 - **Production inference** — sub-millisecond inference, no GPU needed
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3. Feature Engineering (ml/feature_engineering.py)

3.1 Feature Set

Category	Features
Ocean state	SST (°C), SST anomaly (°C from climatology), chlorophyll-a (mg/m³), mixed layer depth (m), D20 isotherm depth (m)
Surface dynamics	Significant wave height (m), wave period (s), swell height/direction, current speed (m/s), current direction (°)
Atmospheric	Wind speed (m/s), wind direction (°), atmospheric pressure (hPa), pressure trend (hPa/3hr), precipitation (mm)
Geospatial	Depth (m, GEBCO), distance from coast (km), distance from nearest port (km), H3 cell ID (resolution 7)
Temporal	Month, day of year, hour, lunar phase (fishing effort correlates with moon)
Derived	SST gradient (spatial), chlorophyll gradient, wind-current alignment, Ekman transport direction

3.2 Feature Processing

```
class FishingFeatureEngine:
    def compute_features(self, ocean: OceanState, weather: WeatherState, geo: GeoState) -> dict:
        features = {}
        features['sst_c'] = ocean.sst_c
        features['sst_anomaly'] = ocean.sst_anomaly_c
        features['chlorophyll_mg_m3'] = ocean.chlorophyll_mg_m3
        features['wave_height_m'] = ocean.significant_wave_height_m
        features['wind_speed_ms'] = weather.wind_speed_ms
        features['depth_m'] = geo.depth_m
        features['month'] = datetime.utcnow().month
        # ... 25+ features total
        return features
```

4. The Label Problem: INCOIS PFZ Archive

4.1 Why PFZ Labels Are Non-Negotiable

The XGBoost model needs training labels — **where fish actually were** on specific dates. The INCOIS PFZ advisory archive (2003–2025) is the only authoritative source of this:

Data	Why It's Needed
PFZ advisory locations	These are the positive labels (fish were here)
PFZ advisory dates	Correlate with ocean conditions on that date
PFZ advisory absence	Negative labels (fish were not predicted here)

Every other dataset in the system is a *feature* (SST, chlorophyll, wind). The PFZ archive is the *label*. Without labels, you cannot train a supervised model.

4.2 Current Status: Blocked

The INCOIS PFZ archive is **not** a bulk public download. Access requires: - MOSDAC portal registration - Institutional affiliation verification - Manual archive request

Status as of Sep 2026: Registration approved, but bulk archive download still pending.

4.3 Synthetic Labels: A Conscious Decision

The ML pipeline includes a synthetic label generator for development purposes: - Uses chlorophyll + SST + depth thresholds to generate “pseudo-PFZ” labels - These are **explicitly** flagged as synthetic throughout the pipeline

Standing rule: > *Synthetic-label metrics are NEVER reported as real results. Every log, every report, every UI display that uses synthetic data says so.*

The pipeline distinguishes real vs. synthetic labels at the data level:

```
df['label_source'] = 'synthetic_v1' # or 'incois_pfz_archive'
```

5. Training Pipeline (ml/train_fishing_model.py)

5.1 Architecture

```
# Pipeline stages
1. Load features (processed Parquet from data/processed/)
2. Load labels (PFZ archive or synthetic)
3. Temporal train/test split (by year, not random – prevents look-ahead bias)
4. Feature selection (correlation + importance)
5. Hyperparameter tuning (Optuna, 100 trials)
6. Train XGBoost model
7. Evaluate on held-out test year
8. Log to MLflow (parameters, metrics, model artifact)
9. Save model to data/models/
```

5.2 Validation Strategy

- **Temporal split:** Train on years 2003–2022, test on 2023–2025
- **Cross-year holdout:** No data from test years leaks into training
- **Metrics:** AUC-ROC, precision, recall, F1 at optimal threshold
- **Backtesting:** How well would historical recommendations have matched actual PFZ advisories?

5.3 MLflow Tracking

Every training run is logged to MLflow: - Parameters: learning_rate, max_depth, n_estimators, feature set version - Metrics: AUC, precision, recall, F1 - Artifacts: Trained model, feature importance plot, confusion matrix - Tags: label_source=synthetic or label_source=incois_pfz

6. Fishing Agent: Current State

6.1 What Works

- Feature engineering pipeline: ✓ complete
- XGBoost training script: ✓ complete
- Inference endpoint wired into the Risk Agent: ✓ ready

6.2 What's Blocked

- **Training on real labels:** Waiting for INCOIS PFZ archive
- **MOSDAC EOS-06 chlorophyll integration:** Adapter exists but needs verified data paths

- **NASA OceanColor chlorophyll fallback:** Not yet wired

6.3 Honest System State

The system currently ranks zones by safety only. It does not claim to find “productive” fishing zones.

The Route Agent’s own documentation says so. This is stated plainly because it affects what may be claimed to judges.

7. Supplementary Data Sources for Fishing Intelligence

Source	Purpose	Status
Global Fishing Watch (GFW)	Validate PFZ against actual historical fishing effort	✓ API key obtained
GBIF/OBIS	Species occurrence by location/season	✓ Public API
BGC-Argo	Subsurface dissolved oxygen, nitrate profiles	✓ Public API
Marine Heatwave layer	Derived from OISST — “why has productivity declined?”	☐ Not yet computed

8. Risk Assessment

Risk	Impact	Mitigation
PFZ archive never becomes available	HIGH — no real labels	Pivot to GFW fishing effort as proxy labels
ML model overfits to training years	MEDIUM	Temporal split validation, cross-year holdout
Synthetic labels contaminate reports	HIGH	Loud labelling at data level + UI level
Feature drift (ocean conditions change)	LOW-MEDIUM	Retrain quarterly; monitor feature distributions

Report prepared by Team ORCA, KIIT University, Bhubaneswar